1137-5017 [illegible] (501)

13. The data is selected from normal population, because the data are very close to straight line.

(502)

20. $H_0: \mu = 74$

$H_a: \mu \neq 74$

$$t = \frac{69.7 - 74}{\frac{16}{\sqrt{25}}} = -1.34375$$

$-t_{24, 0.025} = -2.064$

$\therefore$ Non-reject H_0, because still not too far than $-t_{24, 0.025}$.

30. a) $H_0: \mu = 25$

$H_a: \mu < 25$

$$t = \frac{20.7 - 25}{5/\sqrt{21}} = -4.3076$$

$-t_{20, 0.05} = -1.725$

$\therefore$ reject H_0, ~~$t < t_{20, 0.05}$~~, because t is small than $-t_{20, 0.05}$.

b) 95% CI : $26.6 \pm t_{20, 0.025} \left(\frac{7.4}{\sqrt{21}}\right)$

$= 26.6 \pm 2.086 \left(\frac{7.4}{\sqrt{21}}\right)$

$= [23.2315, 29.9685]$

c) self esteem increase after treatment, follow up remain same.

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55) a) $H_0: \mu = 22$

$H_a: \mu > 22$

$$t = \frac{31-22}{6.2/\sqrt{17}} = 5.9852 > t_{16,\,0.05} = 1.746$$

$\therefore$ reject H_0

b) 90% CI $\Rightarrow$ $31 \pm t_{16,0.025} \times \frac{6.2}{\sqrt{17}}$

$$= 31 \pm 1.746\left(\frac{6.2}{\sqrt{17}}\right)$$

$$= [28.3745,\ 33.6255]$$

(503)

$H_0: \mu_1 - \mu_2 = 0$

$H_a: \mu_1 - \mu_2 \neq 0$

7. $\bar{x}_1 = 7$ $\bar{x}_2 = 8.6$, $s_2^2 = 10.13$ $df = 4+5-2 = 7$

$$S_p^2 = \frac{(4-1)\times 15.33 + (5-1)\times 10.3}{(5+4-2)}$$

$$= 12.46$$

$$t = \frac{7-8.6}{\sqrt{12.4716(\frac{1}{4}+\frac{1}{5})}} \approx -0.6758 > -t_{7,0.025} = 2.365$$

RR $= |t| > t_{7,0.025} = 2.365$

p-value: $2 \times P[t < -0.6758] \approx 0.52 > 0.05$

$\therefore$ no reject, because p-value $0.52 > 0.05$.

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8. $S_p^2 = \dfrac{(16-1)\times 4.0 + (13-1)\times 5.4}{16+13-2}$

≈ 5.2889

$df = 16+13-2$
$= 27$

$t = \dfrac{74.6-32.2}{\sqrt{5.2889\left(\frac{1}{16}+\frac{1}{13}\right)}} \approx 2.7449 > t_{27,0.025} = 2.052$

RR : $|t| > t_{27,0.025} = 2.052$

p-value : $2\times P(t > 2.7449) = 0.0044 < 0.05$

$\therefore$ reject H_0.

12) a) $H_0: \mu_1 - \mu_2 = 0$

$H_a: \mu_1 - \mu_2 \neq 0$

$s = \sqrt{\dfrac{9.45^2}{10} + \dfrac{12.7^2}{10}}$

≈ 4.4113

$t = \dfrac{96.4-74.7}{4.4113} \approx 1.5188 < t_{18,0.025} = 2.101$

$\therefore$ no reject

b) 95% CI : $96.4 \pm t_{18,0.025}(4.4113)$

$= [-2.5681, 15.9681]$

c) because we don't reject H_0 in 95% CI, so we can't say which one is better.

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